

사이클별 지표 패널

M02Ch105

각 패널 = 한 지표의 cycle 궤적 (routine 0.5C).

early/late 점선 = 앞·뒤 5포인트 median.

트렌드 = 선형 slope/100cyc + aging 기대방향 대조.

주의: LAM_PE / *_proxy / pattern_score는 가설·proxy이며 절대 LAM%/LLI%가 아닙니다. R_ohmic은 $\sqrt{t}$ 외삽 proxy (0.1s 단일값 $\neq$ 순수 음).

트렌드 요약 — M02Ch105

- 유효 지표: 48개 · aging 방향 일치 17 · 반대 1

하락 트렌드 (상위)

- LAM 곡선 proxy: early=-3.194 → late=-16.44 (Δ =-13.24, -5.251%/100cyc) → decreasing · context
- LLI 곡선 proxy: early=-0.5851 → late=-2.774 (Δ =-2.189, -1.046%/100cyc) → decreasing · context
- 방전 플래토 폭: early=17.32 → late=15.3 (Δ =-2.017, -0.9287Q-units/100cyc) → decreasing · matches_aging
- R Ω 성장률 /100cyc: early=0.7607 → late=0.4498 (Δ =-0.3109, -0.3132m Ω /100cyc/100cyc) → decreasing · context
- 방전 플래토 ΔV : early=-0.0394 → late=-0.08082 (Δ =-0.04142, -0.01907V/100cyc) → decreasing · context

상승 트렌드 (상위)

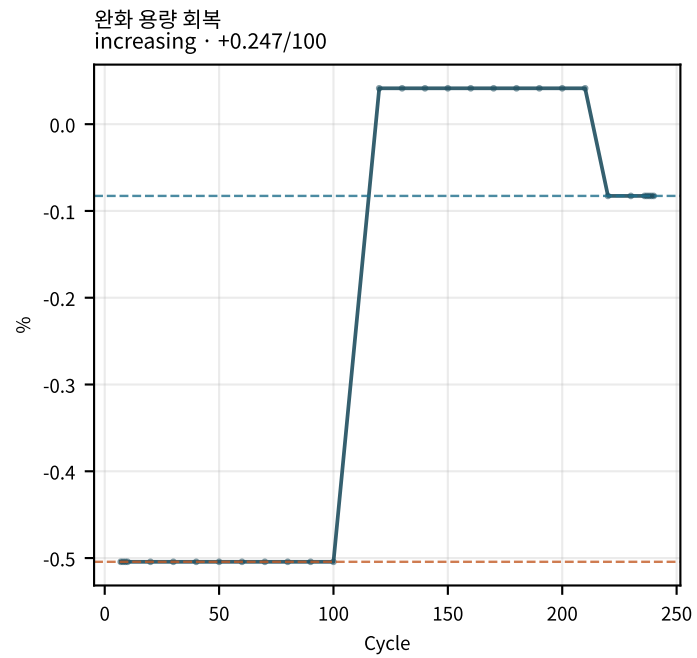
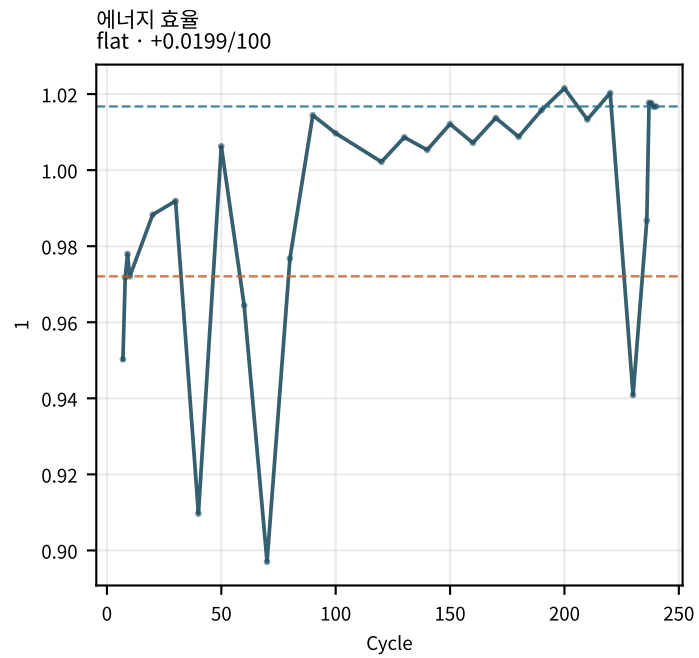
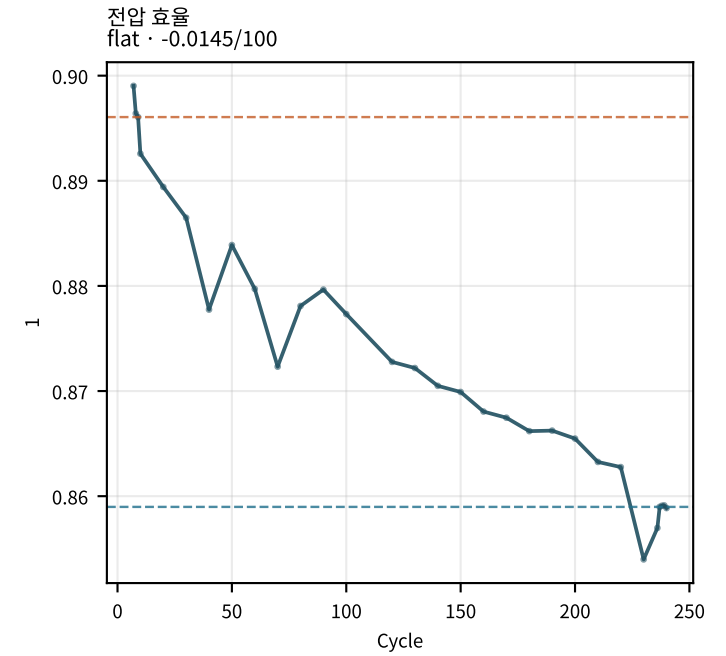
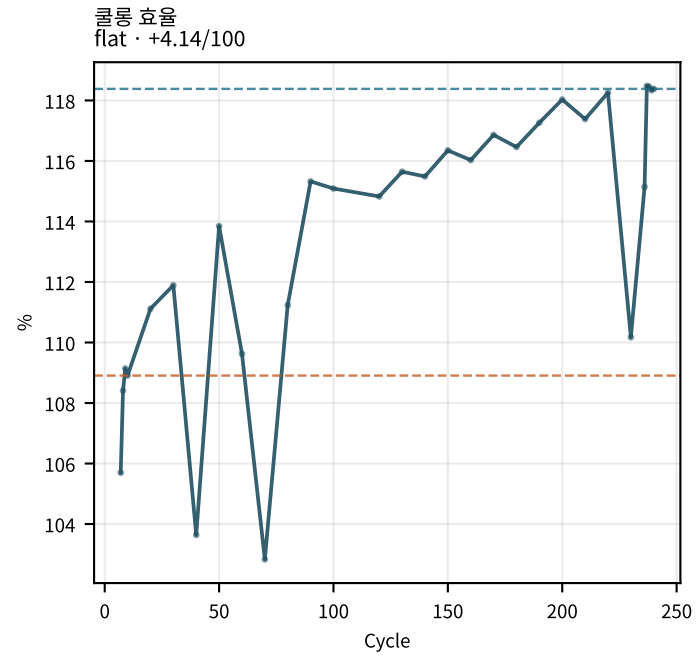
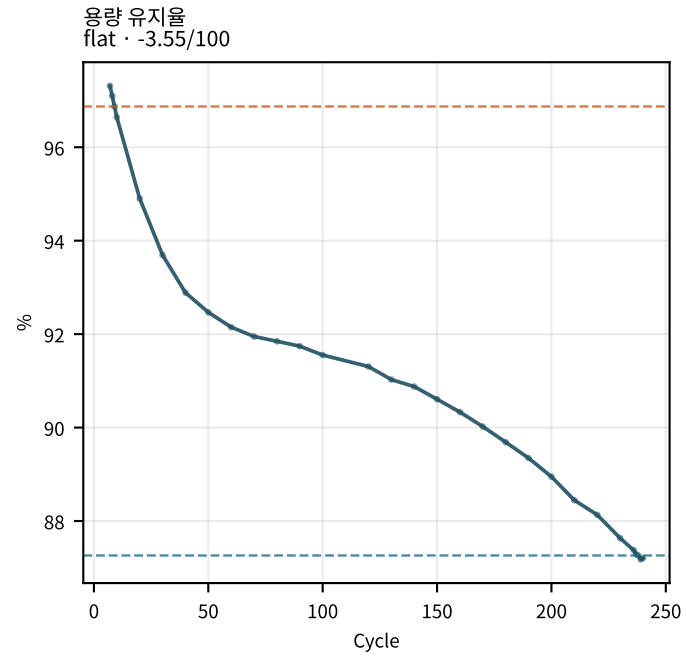
- 음 저항 (SOC50): early=1.349 → late=2.293 (Δ =+0.9446, +0.4789m Ω /100cyc) → increasing · matches_aging
- 기계/화학 비: early=1.86 → late=2.422 (Δ =+0.5618, +0.33311/100cyc) → increasing · matches_aging
- 완화 용량 회복: early=-0.5043 → late=-0.08267 (Δ =+0.4216, +0.247%/100cyc) → increasing · matches_aging
- 계면 R 패턴: early=0.4692 → late=0.8449 (Δ =+0.3757, +0.15480-1/100cyc) → increasing · matches_aging
- PE activity 패턴: early=0.6565 → late=0.9319 (Δ =+0.2754, +0.1380-1/100cyc) → increasing · matches_aging

CycleDiag : plot_cycle_metrics_panel.py

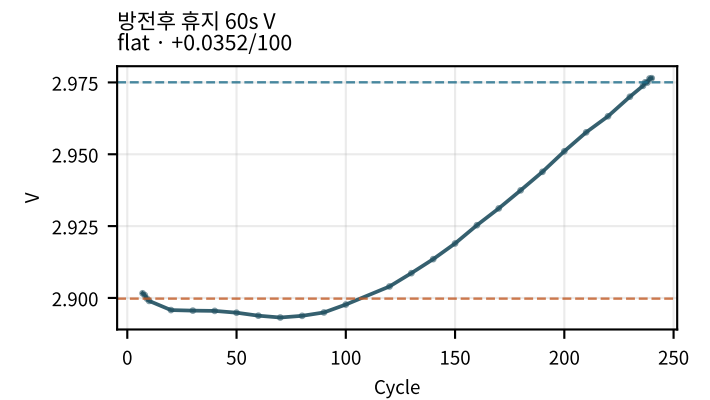
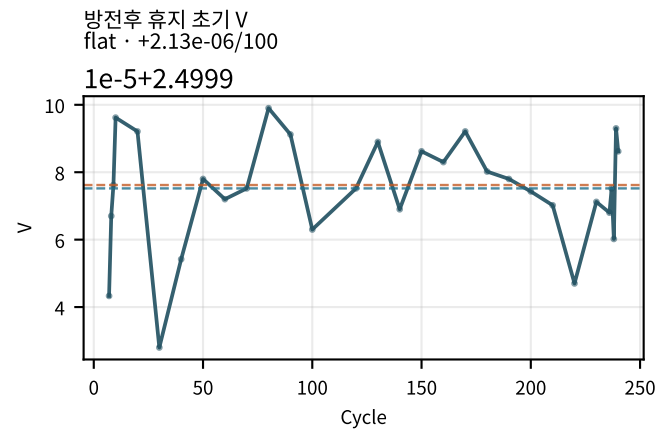
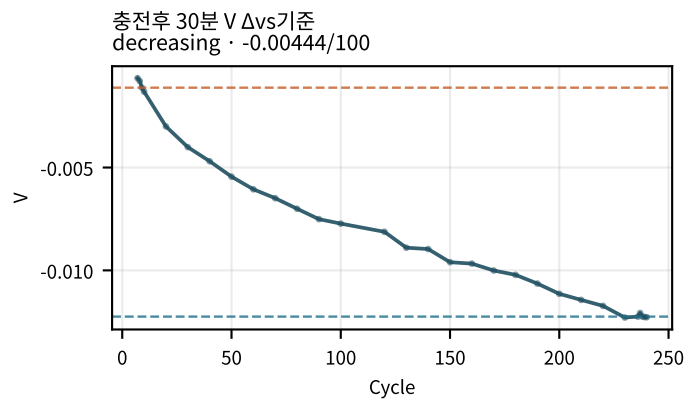
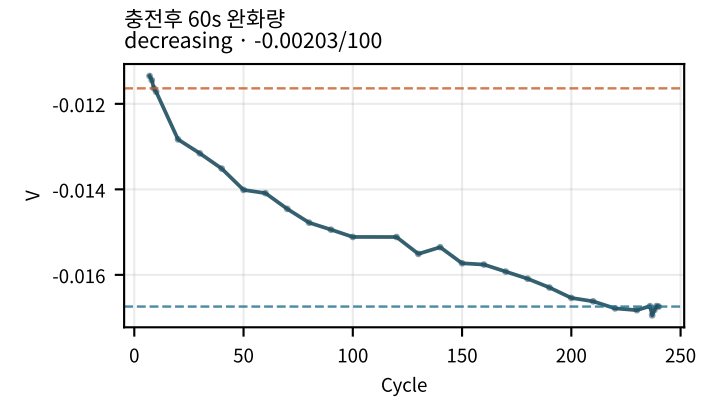
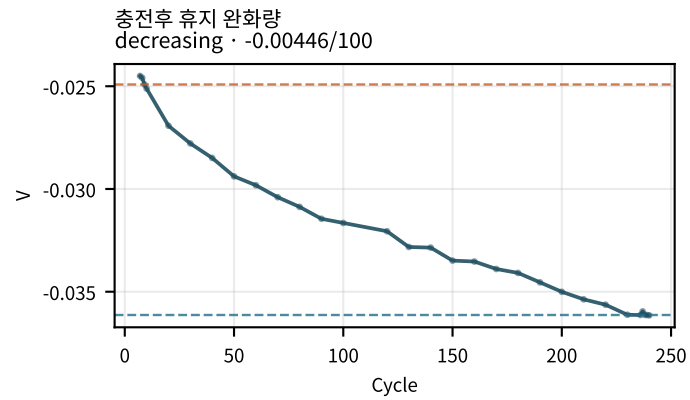
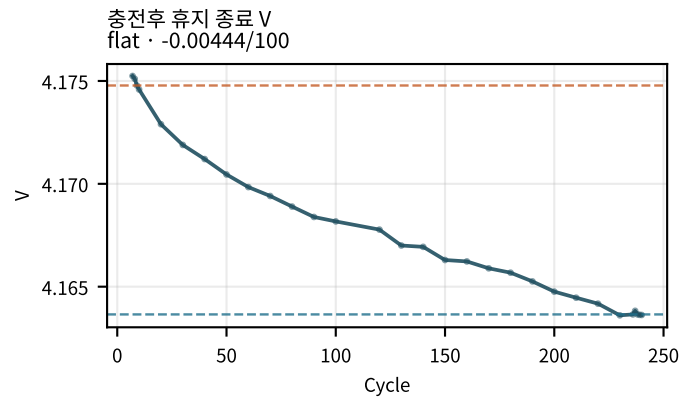
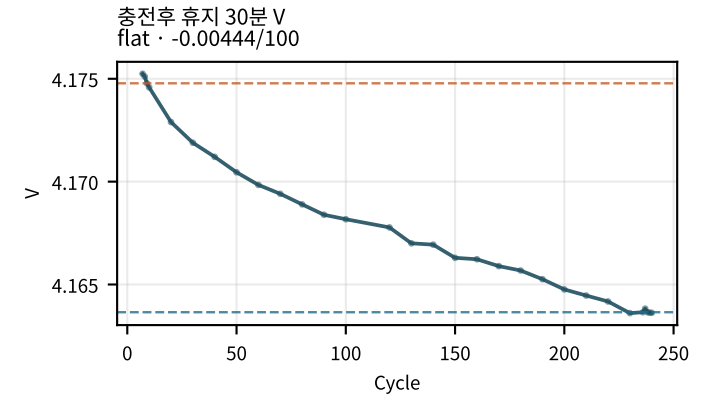
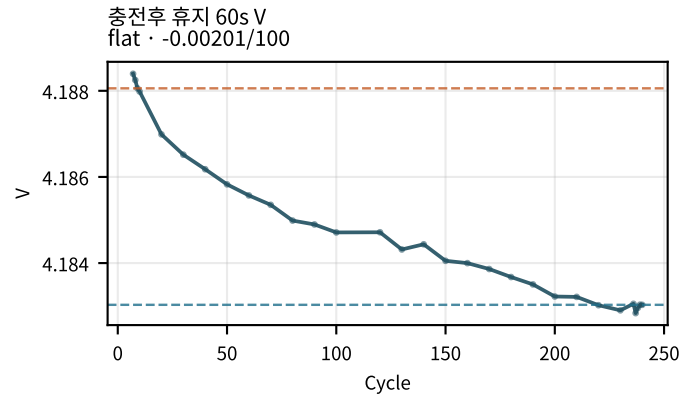
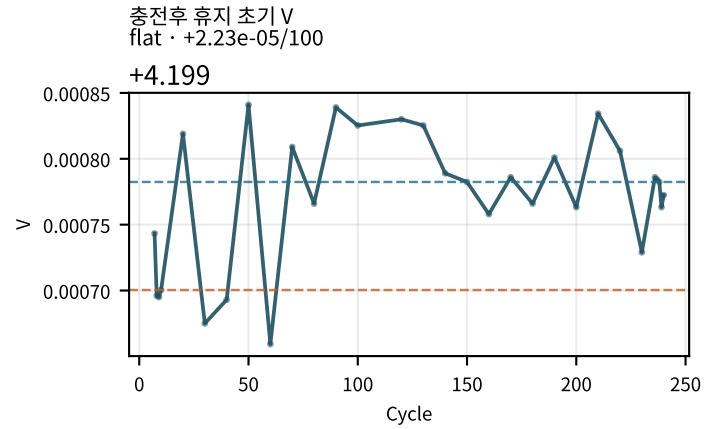
기대 aging과 반대

- 저SOC 히스테리시스: early=0.07667 → late=0.04561 (Δ =-0.03106, -0.01231V/100cyc) → decreasing · opposite_aging

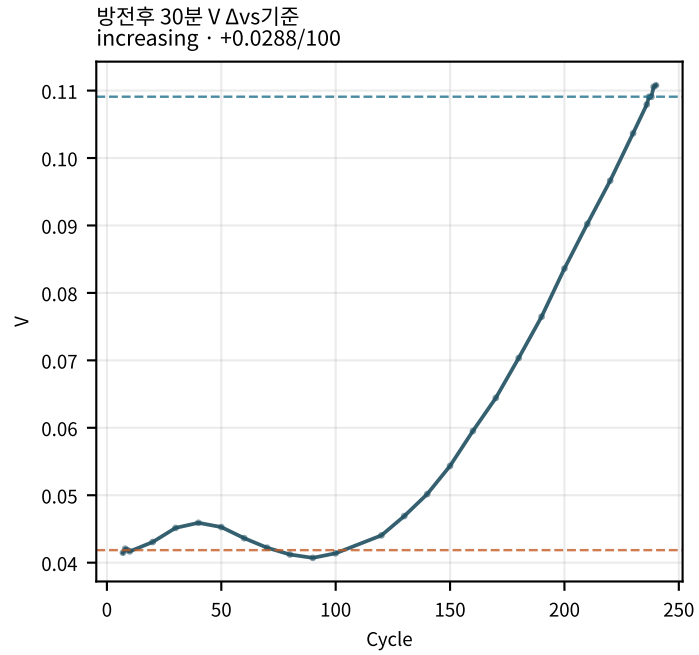
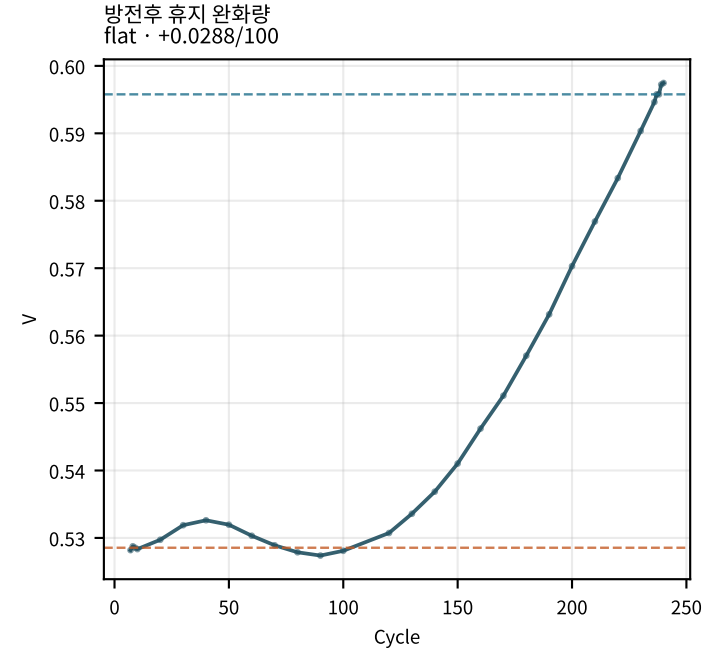
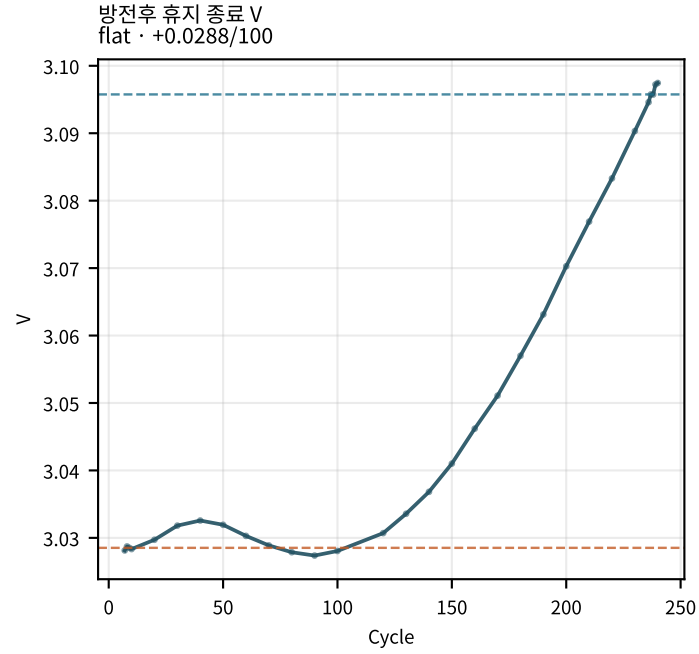
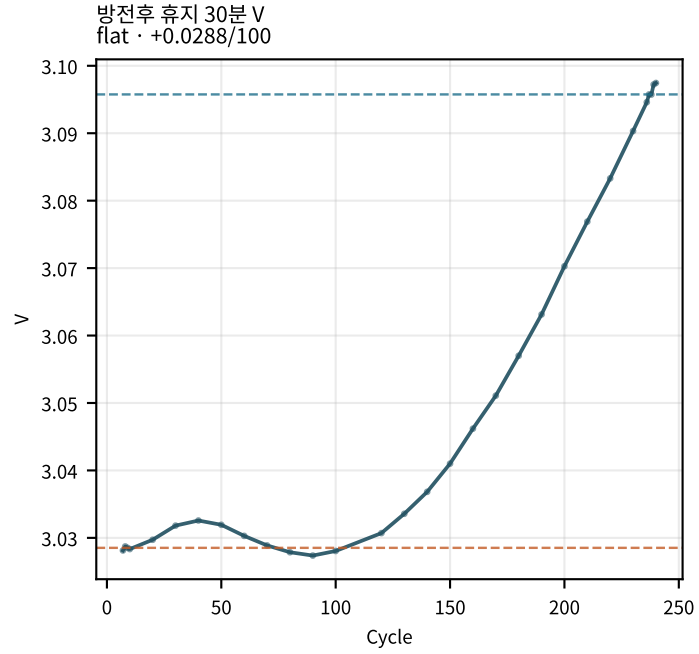
M02Ch105 · 용량 · 효율



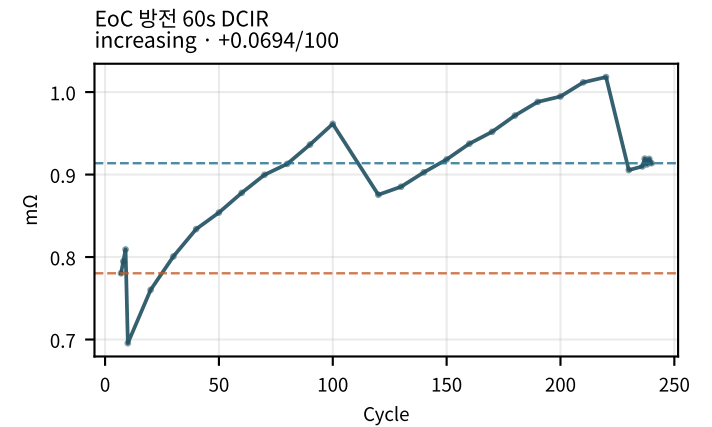
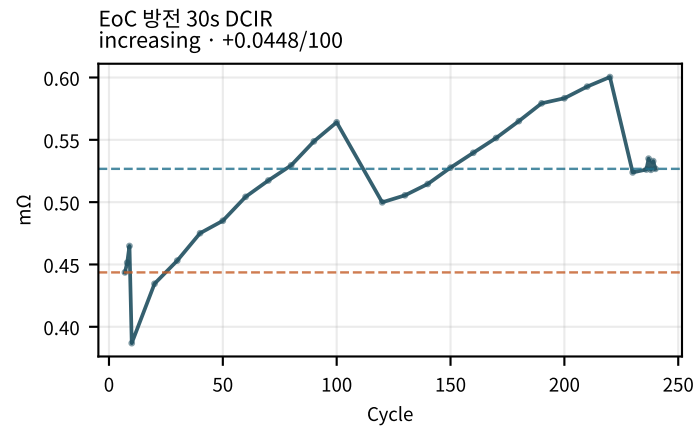
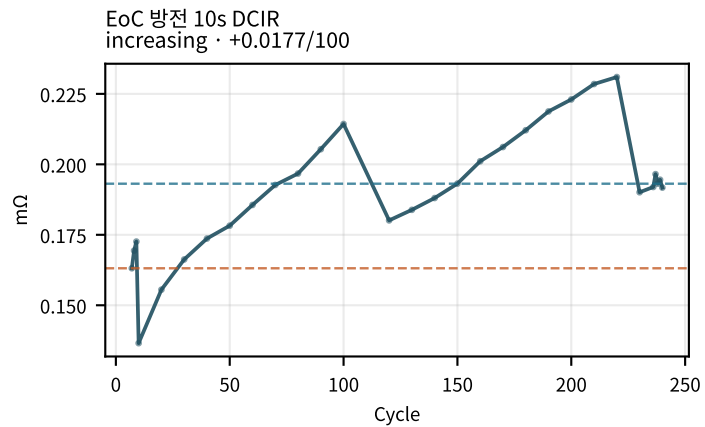
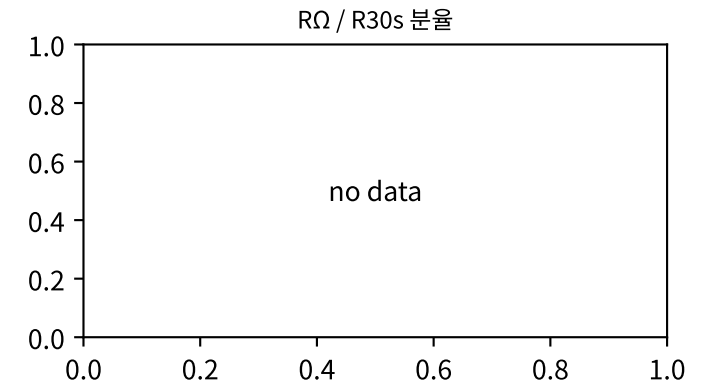
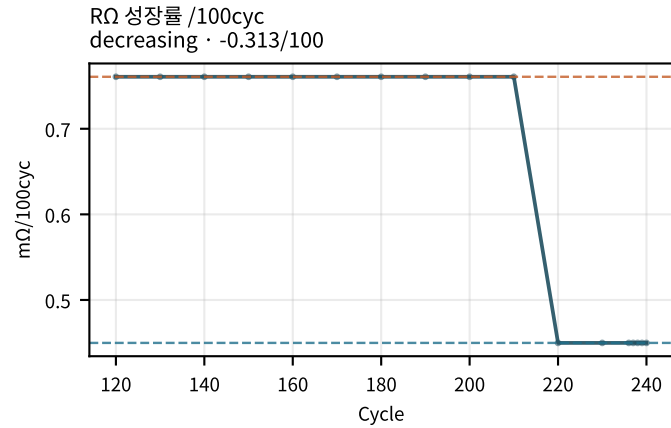
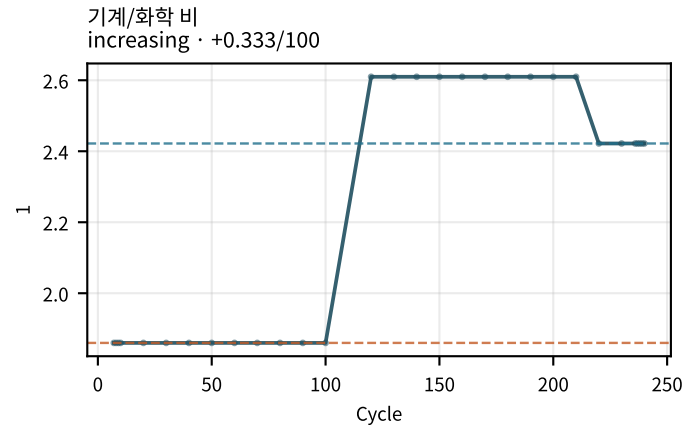
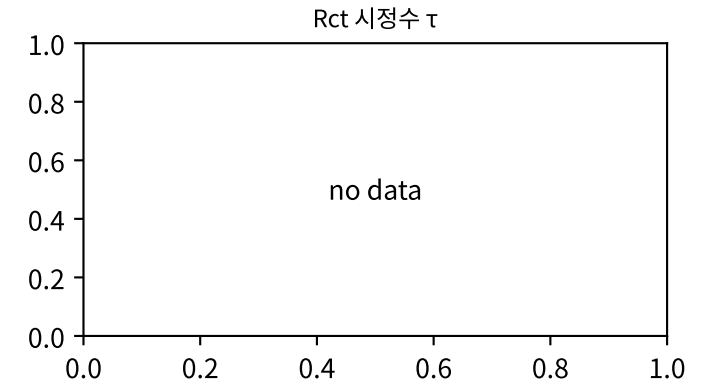
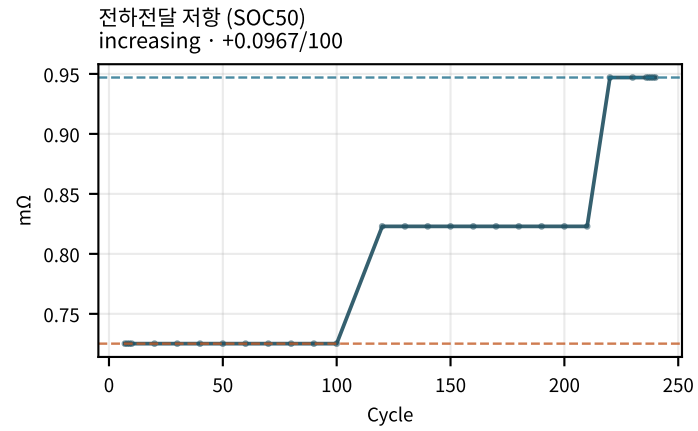
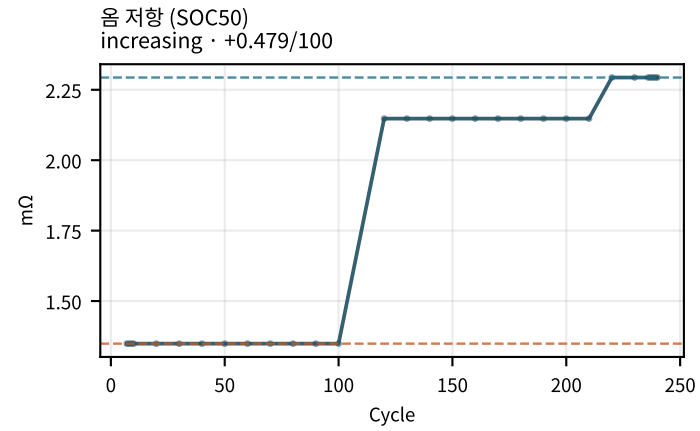
M02Ch105 · 휴지 전압 (충전/방전 후)



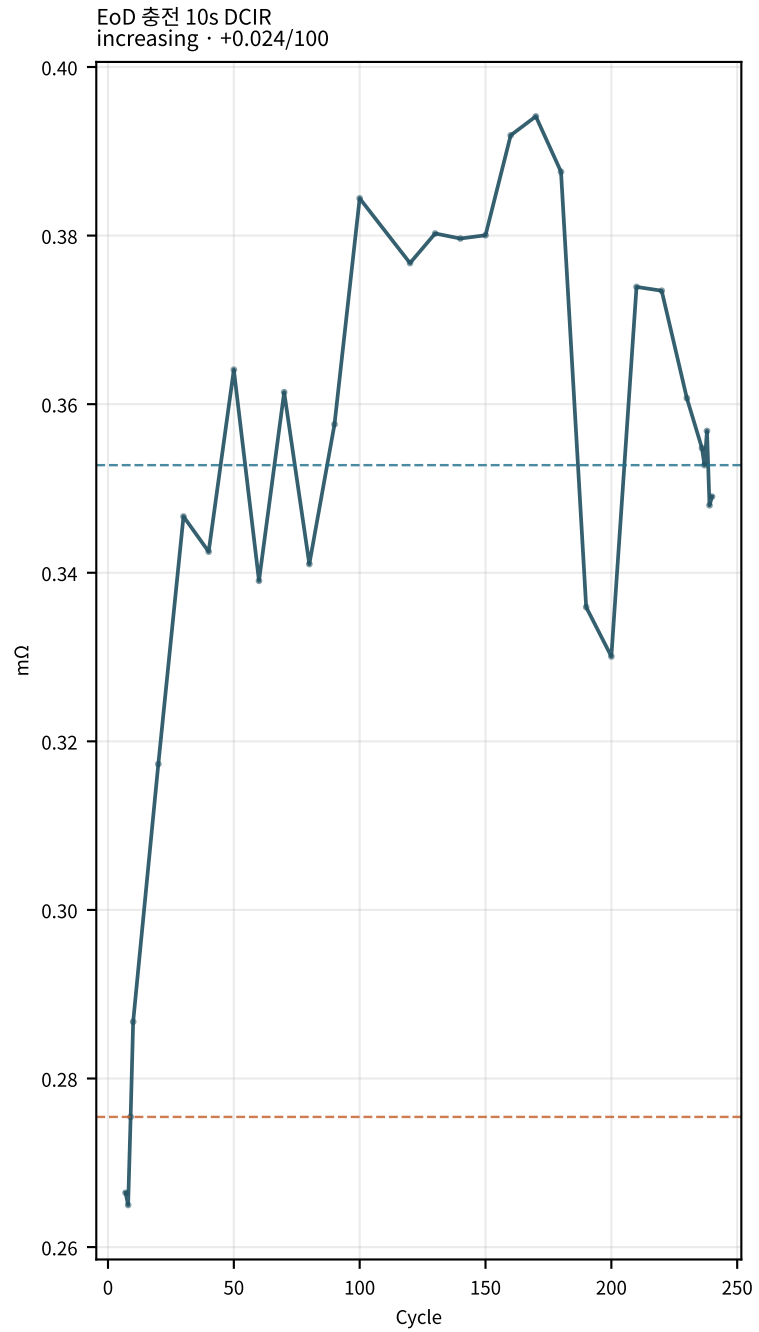
M02Ch105 · 휴지 전압 (충전/방전 후)



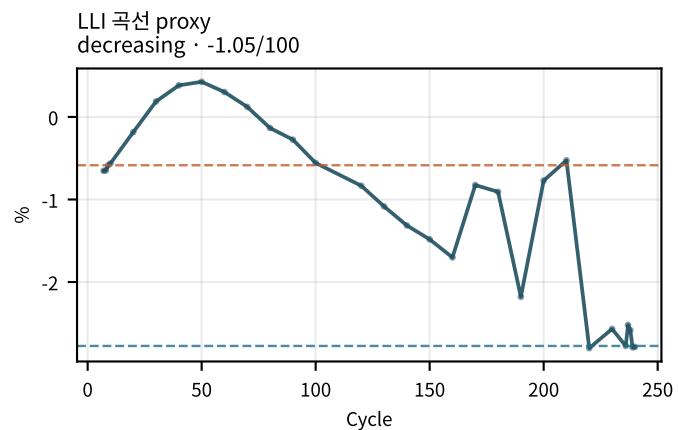
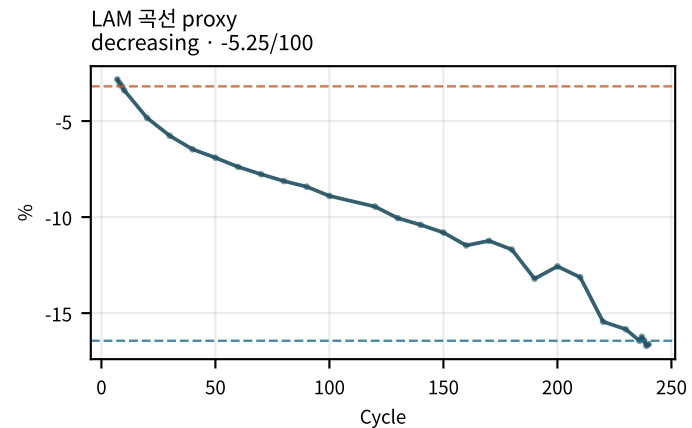
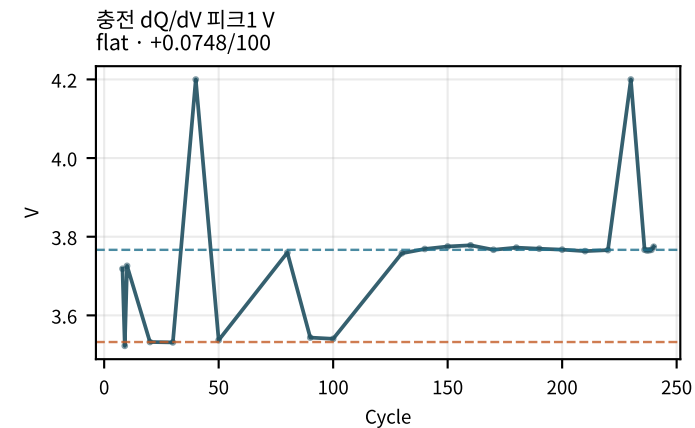
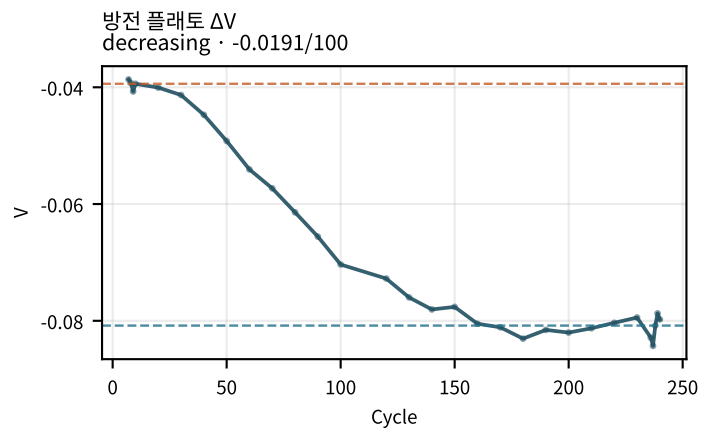
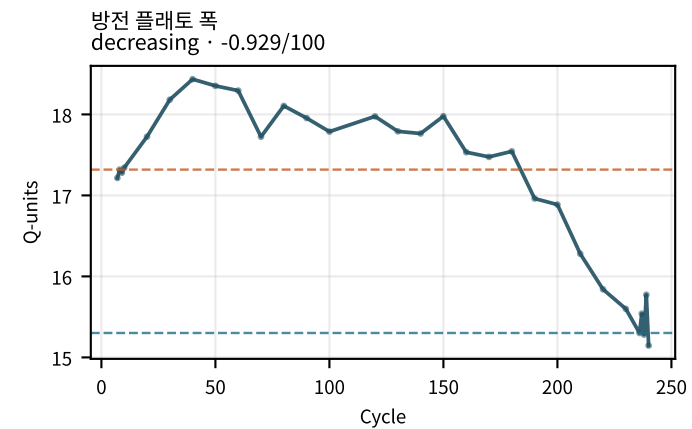
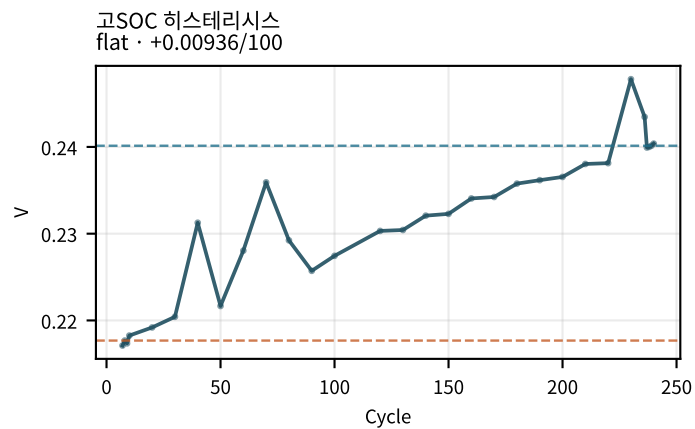
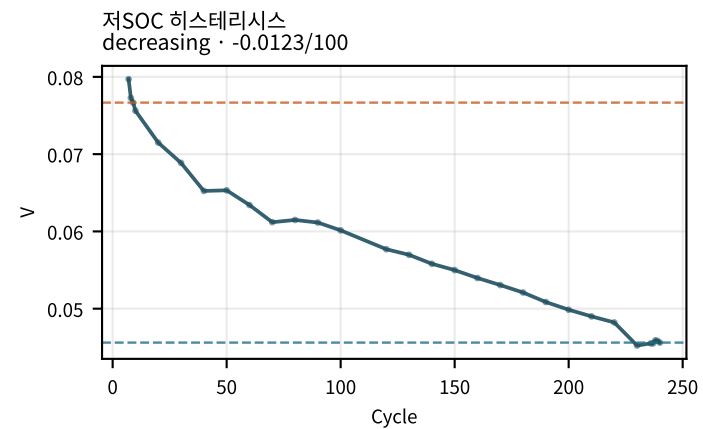
M02Ch105 · 저항 · 분해



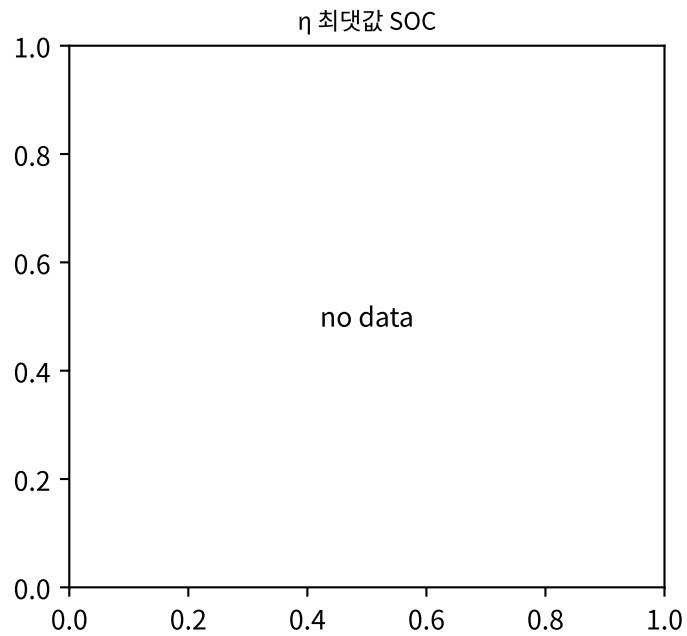
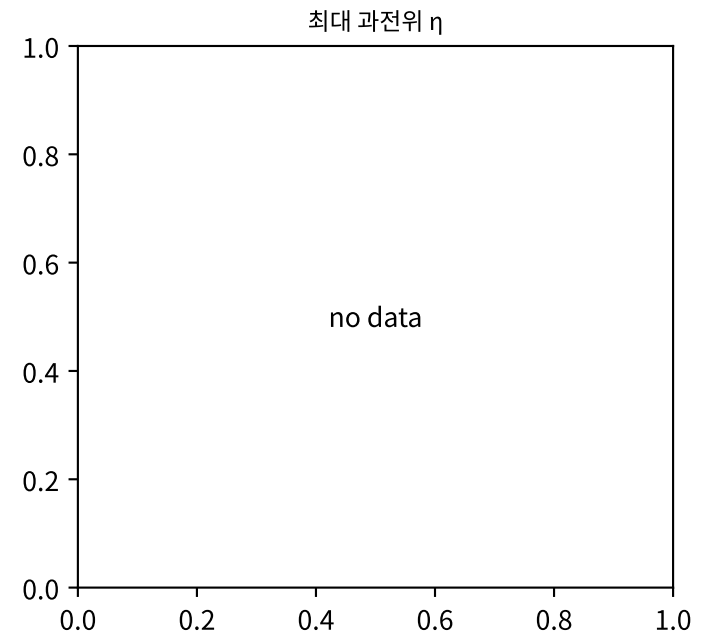
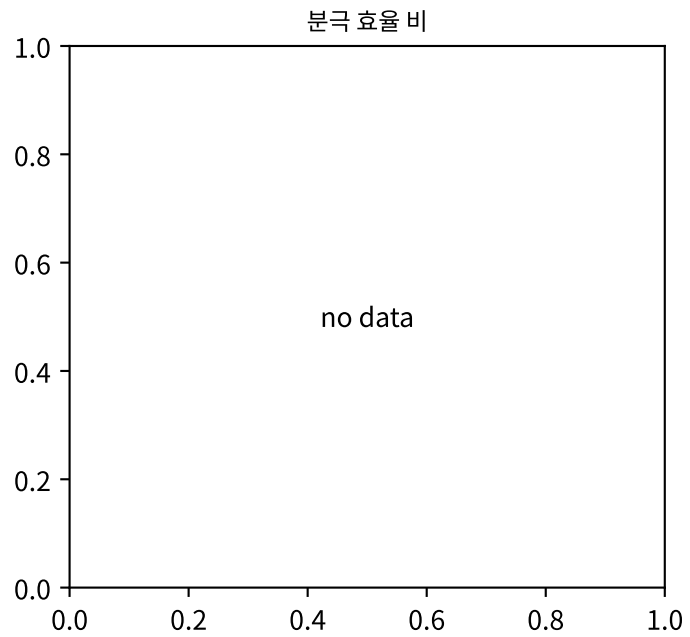
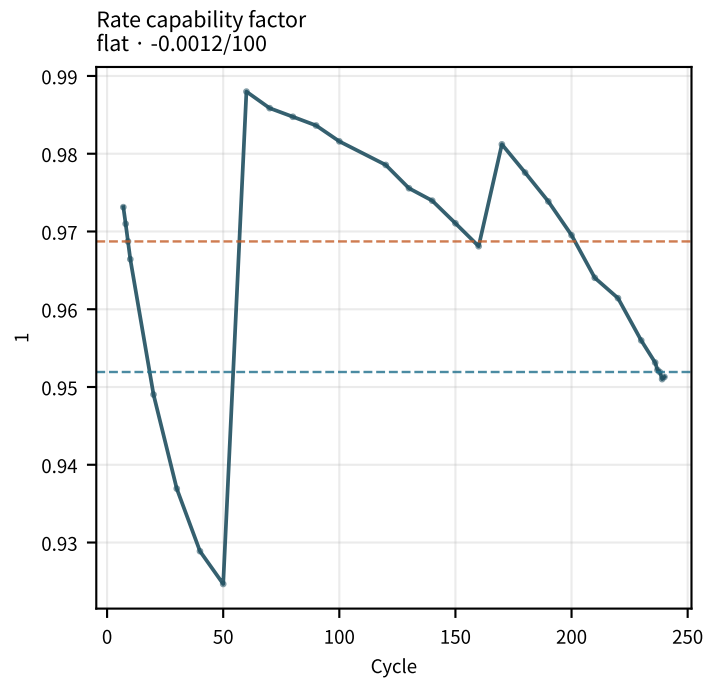
M02Ch105 · 저항 · 분해



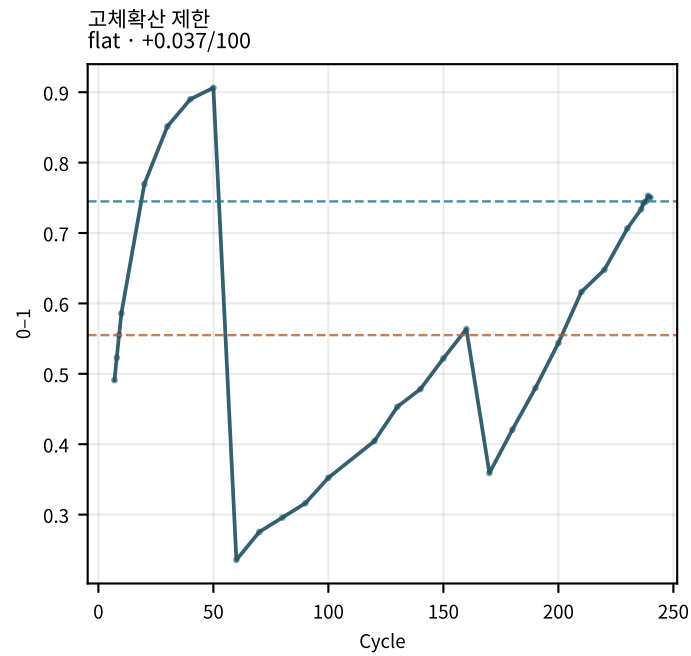
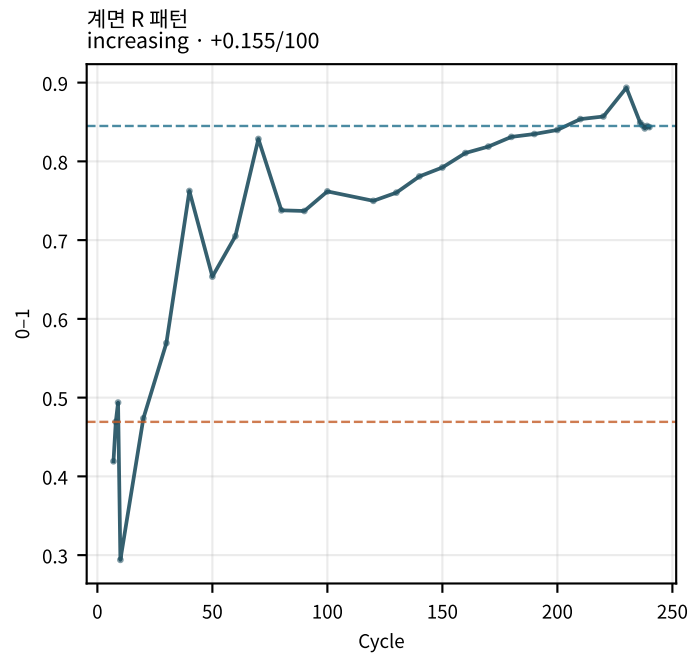
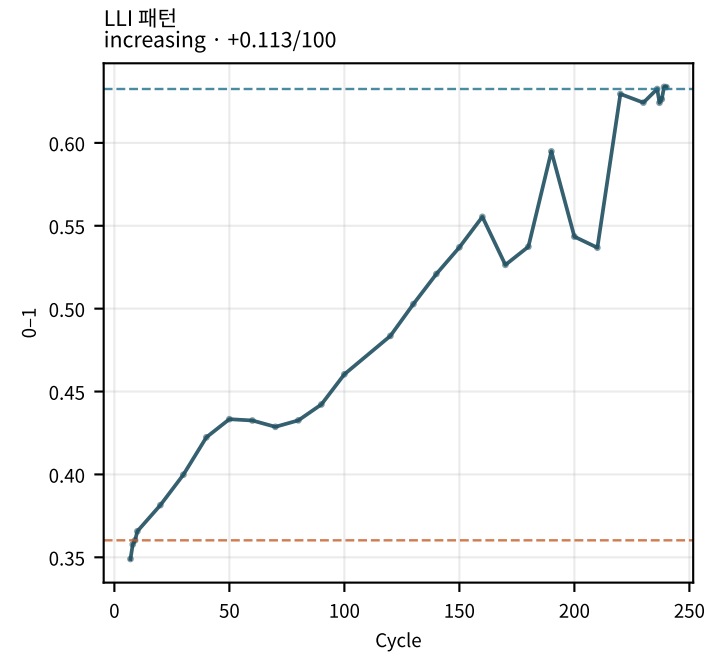
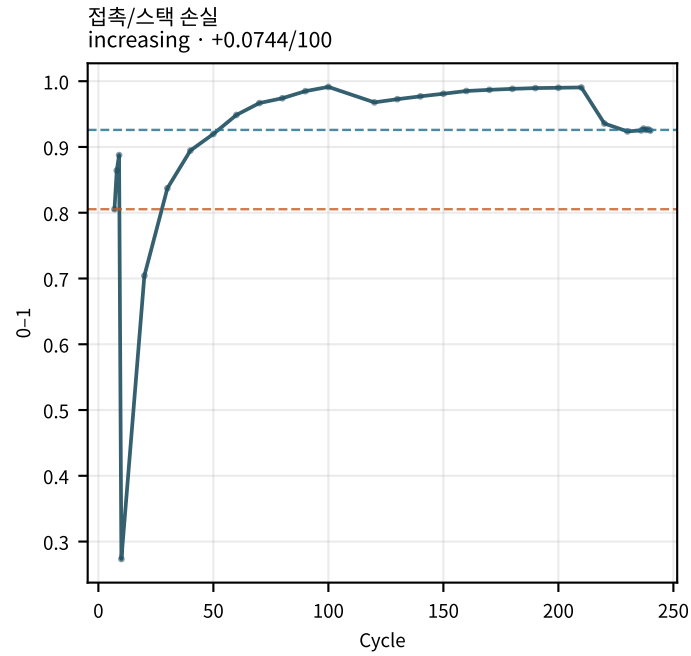
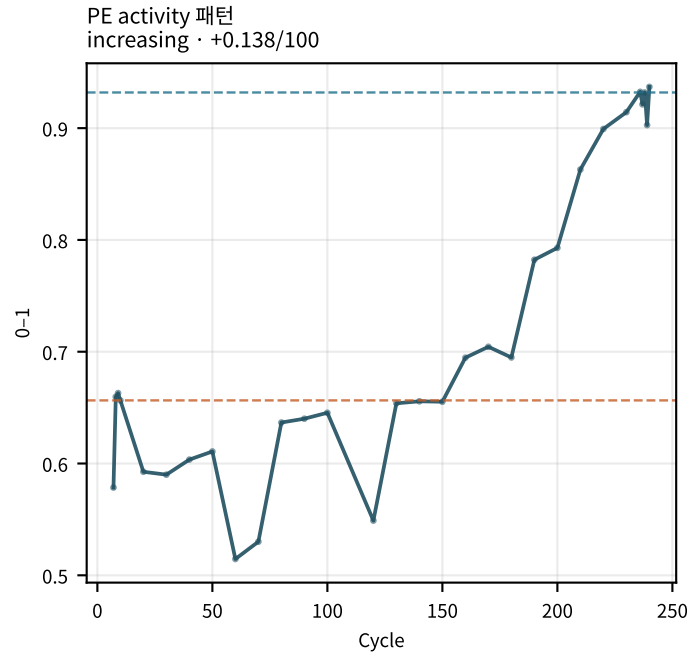
M02Ch105 · 곡선 형상 · dQ/dV



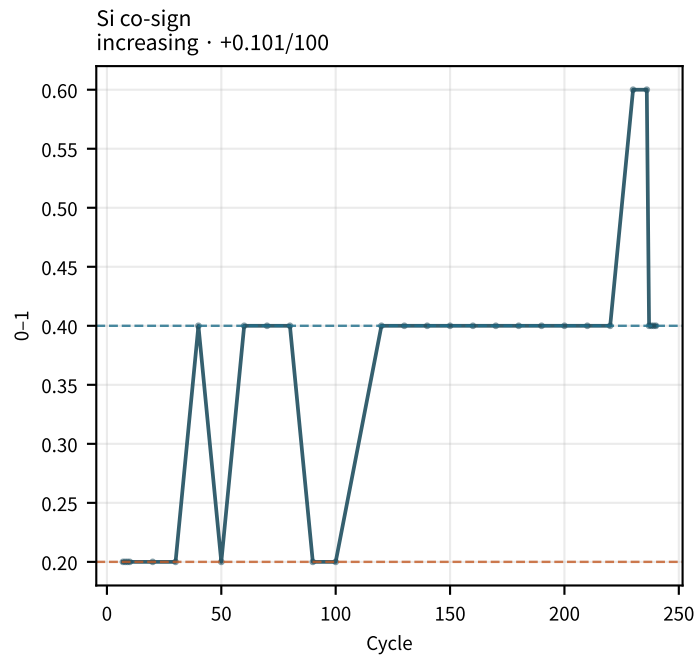
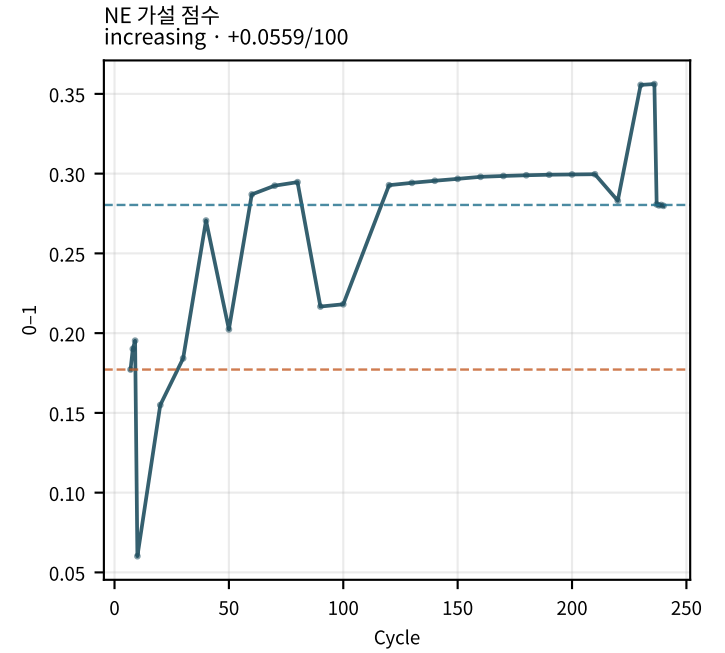
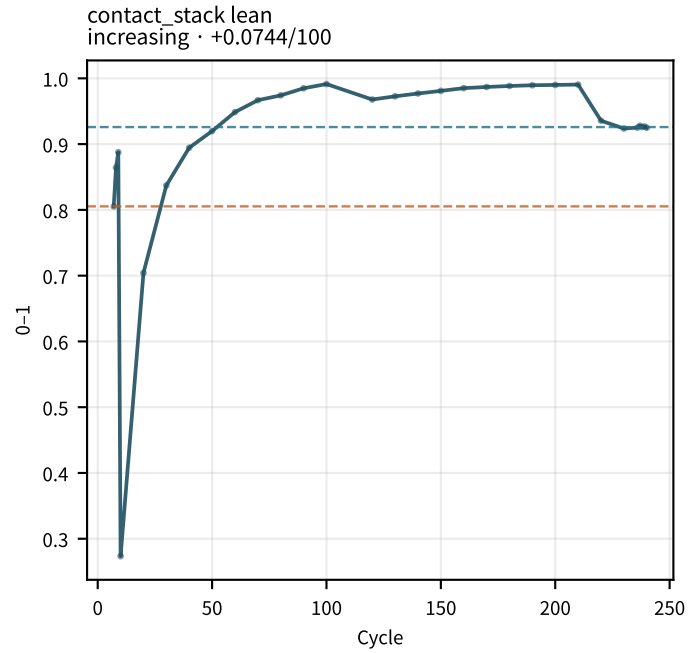
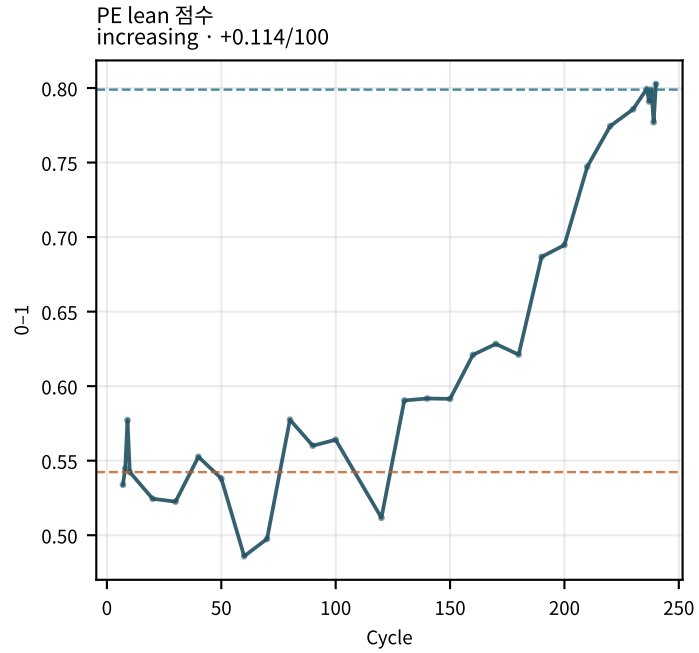
M02Ch105 · 수송 · rate



M02Ch105 · 열화 패턴 점수



M02Ch105 · 전극 lean 가설



지표 설명 · 트렌드 (M02Ch105)

용량 유지율 (SoHQ)

기준 대비 방전용량 유지율. fade/knee의 1차 관측량. | 계산: $Q_dchg(cycle) / Q_baseline \times 100$ (routine 0.5C).

트렌드: 용량 유지율: early=96.87 → late=87.26 ($\Delta=-9.61$, -3.55%/100cyc) → flat · stable | aging 기대=decrease | vs=stable

쿨롱 효율 (CE)

충전 대비 방전용량 비. LLI·부반응 힌트. | 계산: $Q_dchg / Q_chg \times 100$ (이상치 >102% 또는 <85%는 진단에서 제외).

트렌드: 쿨롱 효율: early=108.9 → late=118.4 ($\Delta=+9.475$, +4.139%/100cyc) → flat · stable | aging 기대=decrease | vs=stable

전압 효율 (VE)

에너지 효율의 전압 측. 저항·분극 증가 시 하락. | 계산: E_dchg / E_chg (Wh).

트렌드: 전압 효율: early=0.8961 → late=0.859 ($\Delta=-0.03708$, -0.014471/100cyc) → flat · stable | aging 기대=decrease | vs=stable

에너지 효율 (EE)

충·방전 에너지 비. | 계산: E_dchg / E_chg .

트렌드: 에너지 효율: early=0.9721 → late=1.017 ($\Delta=+0.04467$, +0.019871/100cyc) → flat · stable | aging 기대=decrease | vs=stable

완화 용량 회복 (Q_relax_pct)

DCIR/RPT 전후 휴지로 회복되는 용량 분율. Si chemo-mech co-sign. | 계산: DCIR 블록 전후 Q 차이 / Q × 100, routine에 forward-fill.

트렌드: 완화 용량 회복: early=-0.5043 → late=-0.08267 ($\Delta=+0.4216$, +0.247%/100cyc) → increasing · matches_aging | aging 기대=increase | vs=matches_aging

충전후 휴지 초기 V (EoC_restV_init)

충전(EoC) 직후 휴지 시작 전압. | 계산: charge 종료 후 rest step 첫 샘플 전압.

트렌드: 충전후 휴지 초기 V: early=4.2 → late=4.2 ($\Delta=+8.2e-05$, +2.233e-05V/100cyc) → flat · context | aging 기대=either | vs=context

지표 설명 · 트렌드 (M02Ch105)

충전후 휴지 60s V (EoC_restV_60s)

충전 후 휴지 60초 시점 전압. | 계산: rest step_time $\approx$ 60 s 보간.

트렌드: 충전후 휴지 60s V: early=4.188 $\rightarrow$ late=4.183 (Δ =-0.005026, -0.002008V/100cyc) $\rightarrow$ flat · context | aging
기대=either | vs=context

충전후 휴지 30분 V (EoC_restV_30m)

충전 후 휴지 30분 시점 전압. OCV에 가까운 EoC rest. | 계산: rest step_time $\approx$ 1800 s 보간.

트렌드: 충전후 휴지 30분 V: early=4.175 $\rightarrow$ late=4.164 (Δ =-0.01113, -0.004441V/100cyc) $\rightarrow$ flat · context | aging
기대=either | vs=context

충전후 휴지 종료 V (EoC_restV_end)

충전 후 휴지 스텝 마지막 전압. | 계산: rest step 끝 샘플.

트렌드: 충전후 휴지 종료 V: early=4.175 $\rightarrow$ late=4.164 (Δ =-0.01113, -0.004441V/100cyc) $\rightarrow$ flat · context | aging
기대=either | vs=context

충전후 휴지 완화량 (EoC_restV_relax)

휴지 동안 전압 변화 (end – init). 분극 완화. | 계산: EoC_restV_end – EoC_restV_init.

트렌드: 충전후 휴지 완화량: early=-0.02491 $\rightarrow$ late=-0.03613 (Δ =-0.01122, -0.004464V/100cyc) $\rightarrow$ decreasing · context |
aging 기대=either | vs=context

충전후 60s 완화량 (EoC_restV_relax_60s)

휴지 첫 60초 완화량. | 계산: EoC_restV_60s – EoC_restV_init.

트렌드: 충전후 60s 완화량: early=-0.01164 $\rightarrow$ late=-0.01674 (Δ =-0.005104, -0.00203V/100cyc) $\rightarrow$ decreasing · context |
aging 기대=either | vs=context

충전후 30분 V Δ vs기준 (delta_EoC_restV_30m)

기준 사이클 대비 충전후 30분 rest V 이동 (slippage/LLI 힌트). | 계산: EoC_restV_30m(cycle) – EoC_restV_30m(baseline).

트렌드: 충전후 30분 V Δ vs기준: early=-0.001119 $\rightarrow$ late=-0.01225 (Δ =-0.01113, -0.004441V/100cyc) $\rightarrow$ decreasing · context
| aging 기대=either | vs=context

지표 설명 · 트렌드 (M02Ch105)

방전후 휴지 초기 V (EoD_restV_init)

방전(EoD) 직후 휴지 시작 전압. | 계산: discharge 종료 후 rest step 첫 샘플.

트렌드: 방전후 휴지 초기 V: early=2.5 → late=2.5 ($\Delta=-1\text{e-}06$, $+2.133\text{e-}06\text{V}/100\text{cyc}$) → flat · context | aging 기대=either | vs=context

방전후 휴지 60s V (EoD_restV_60s)

방전 후 휴지 60초 전압. | 계산: rest $\approx$ 60 s 보간.

트렌드: 방전후 휴지 60s V: early=2.9 → late=2.975 ($\Delta=+0.07528$, $+0.03516\text{V}/100\text{cyc}$) → flat · context | aging 기대=either | vs=context

방전후 휴지 30분 V (EoD_restV_30m)

방전 후 휴지 30분 전압. OCV에 가까운 EoD rest. | 계산: rest $\approx$ 1800 s 보간.

트렌드: 방전후 휴지 30분 V: early=3.029 → late=3.096 ($\Delta=+0.06723$, $+0.02883\text{V}/100\text{cyc}$) → flat · context | aging 기대=either | vs=context

방전후 휴지 종료 V (EoD_restV_end)

방전 후 휴지 스텝 마지막 전압. | 계산: rest step 끝 샘플.

트렌드: 방전후 휴지 종료 V: early=3.029 → late=3.096 ($\Delta=+0.06723$, $+0.02883\text{V}/100\text{cyc}$) → flat · context | aging 기대=either | vs=context

방전후 휴지 완화량 (EoD_restV_relax)

방전 후 휴지 완화 (end – init). | 계산: EoD_restV_end – EoD_restV_init.

트렌드: 방전후 휴지 완화량: early=0.5285 → late=0.5958 ($\Delta=+0.06724$, $+0.02882\text{V}/100\text{cyc}$) → flat · context | aging 기대=either | vs=context

방전후 30분 V Δ vs기준 (delta_EoD_restV_30m)

기준 대비 방전후 30분 rest V 이동. | 계산: EoD_restV_30m(cycle) – baseline.

트렌드: 방전후 30분 V Δ vs기준: early=0.04186 → late=0.1091 ($\Delta=+0.06723$, $+0.02883\text{V}/100\text{cyc}$) → increasing · context | aging 기대=either | vs=context

지표 설명 · 트렌드 (M02Ch105)

옴 저항 (SOC50) (R_ohmic_soc50)

DCIR 펄스 early $\sqrt{t}$ 외삽 $t \rightarrow 0$ 절편. 순수 옴이 아니라 초기 비옴 성분 proxy. | 계산: $R(t) = |\Delta V|/|I|$; $t \in [0, 0.3]s$ 에서 R vs $\sqrt{t}$ 회귀 절편.
SOC50 블록 $\rightarrow$ forward-fill.

트렌드: 옴 저항 (SOC50): early=1.349 $\rightarrow$ late=2.293 ($\Delta = +0.9446$, $+0.4789m\Omega/100cyc$) $\rightarrow$ increasing · matches_aging |
aging 기대=increase | vs=matches_aging

전하전달 저항 (SOC50) (R_ct_soc50)

중간 시간대 잔차의 지수 포화항. 계면/화학 저항 proxy (Cdl 직접 추정 아님). | 계산: $resid = R - R\Omega - A\sqrt{t}$; $t \in [0.3, 10]s$ 에서 $R_{ct}(1 - e^{-(t/\tau)})$ fit.

트렌드: 전하전달 저항 (SOC50): early=0.7252 $\rightarrow$ late=0.947 ($\Delta = +0.2218$, $+0.09667m\Omega/100cyc$) $\rightarrow$ increasing · matches_aging |
aging 기대=increase | vs=matches_aging

Rct 시정수 τ (tau_ct_soc50)

$R(t)$ 잔차 fit의 유효 시정수. $\tau \approx R_{ct} \cdot C_{dl}$ 가설이지만 Cdl은 분리하지 않음. | 계산: curve_fit 초기값 2s, bounds [0.05, 20]; 데이터로 추정.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

기계/화학 비 (mech_vs_chem_ratio)

옴(스택·접촉) vs 전하전달(계면) 상대 비중. | 계산: $R_{ohmic_soc50} / R_{ct_soc50}$.

트렌드: 기계/화학 비: early=1.86 $\rightarrow$ late=2.422 ($\Delta = +0.5618$, $+0.33311/100cyc$) $\rightarrow$ increasing · matches_aging | aging
기대=increase | vs=matches_aging

R Ω 성장률 /100cyc (R_ohmic_growth_100)

기준 DCIR 대비 100사이클당 옴 저항 증가율. 레벨($R\Omega$)과 별개 — 감속해도 $R\Omega$ 는 계속 상승 가능. | 계산: $(R\Omega - R\Omega_0) / ((cyc - cyc_0)/100)$.

트렌드: R Ω 성장률 /100cyc: early=0.7607 $\rightarrow$ late=0.4498 ($\Delta = -0.3109$, $-0.3132m\Omega/100cyc/100cyc$) $\rightarrow$ decreasing · context |
aging 기대=either | vs=context

R Ω / R30s 분율 (R_ohmic_frac_soc50)

30s 총저항 중 옴 성분 비중. | 계산: $R_{ohmic} / R_{30s_total}$ @ SOC50.

트렌드: 데이터 부족 | aging 기대=increase | vs=n/a

지표 설명 · 트렌드 (M02Ch105)

EoC 방전 10s DCIR (EoC_dchgR_10s)

충전 종료 후 방전 시작 10초 시점 $\Delta V/I$. Rct·확산 포함 총 DCIR. | 계산: $|V0-V(10s)|/|I|*1000$. 순수 R Ω 아님.

트렌드: EoC 방전 10s DCIR: early=0.1631 → late=0.1931 (Δ =+0.03, +0.01766m Ω /100cyc) → increasing · matches_aging | aging 기대=increase | vs=matches_aging

EoC 방전 30s DCIR (EoC_dchgR_30s)

충전 후 방전 30초 시점 DCIR. | 계산: $|V0-V(30s)|/|I|*1000$.

트렌드: EoC 방전 30s DCIR: early=0.4436 → late=0.5267 (Δ =+0.08309, +0.04483m Ω /100cyc) → increasing · matches_aging | aging 기대=increase | vs=matches_aging

EoC 방전 60s DCIR (EoC_dchgR_60s)

충전 후 방전 60초 시점 DCIR. | 계산: $|V0-V(60s)|/|I|*1000$.

트렌드: EoC 방전 60s DCIR: early=0.7803 → late=0.9138 (Δ =+0.1334, +0.06938m Ω /100cyc) → increasing · matches_aging | aging 기대=increase | vs=matches_aging

EoD 충전 10s DCIR (EoD_chgR_10s)

방전 종료 후 충전 시작 10초 DCIR. | 계산: $|V0-V(10s)|/|I|*1000$.

트렌드: EoD 충전 10s DCIR: early=0.2754 → late=0.3528 (Δ =+0.07732, +0.02403m Ω /100cyc) → increasing · matches_aging | aging 기대=increase | vs=matches_aging

저SOC 히스테리시스 (hyst_area_low)

저SOC 충·방전 전압 면적. Si-on-Gr chemo-mech 지표. | 계산: V-Q 히스테리시스를 SOC 밴드별로 적분 (low band).

트렌드: 저SOC 히스테리시스: early=0.07667 → late=0.04561 (Δ =-0.03106, -0.01231V/100cyc) → decreasing · opposite_aging | aging 기대=increase | vs=opposite_aging

고SOC 히스테리시스 (hyst_area_high)

고SOC 히스테리시스. PE activity 보조 증거. | 계산: 히스테리시스 high-SOC band 면적.

트렌드: 고SOC 히스테리시스: early=0.2177 → late=0.2401 (Δ =+0.02246, +0.00936V/100cyc) → flat · stable | aging 기대=increase | vs=stable

지표 설명 · 트렌드 (M02Ch105)

방전 플래토 폭 (dchg_plateau_width)

방전 플래토 구간 폭. 좁아지면 PE isolation/activity 패턴. | 계산: 방전 V 플래토 검출 후 Q 폭.

트렌드: 방전 플래토 폭: early=17.32 → late=15.3 ($\Delta=-2.017$, -0.9287Q-units/100cyc) → decreasing · matches_aging | aging 기대=decrease | vs=matches_aging

방전 플래토 ΔV (delta_dchg_plateau_V)

기준 대비 방전 플래토 전압 이동. | 계산: plateau_V(cycle) - plateau_V(baseline).

트렌드: 방전 플래토 ΔV : early=-0.0394 → late=-0.08082 ($\Delta=-0.04142$, -0.01907V/100cyc) → decreasing · context | aging 기대=either | vs=context

충전 dQ/dV 피크1 V (chg_dQdV_peak1_V)

충전 IC 1번 피크 위치. FC-OCP 매칭·위상 이동 추적. | 계산: Q 보간 → SG → find_peaks.

트렌드: 충전 dQ/dV 피크1 V: early=3.532 → late=3.767 ($\Delta=+0.2344$, +0.07479V/100cyc) → flat · context | aging 기대=either | vs=context

LAM 곡선 proxy (LAM_curve_proxy)

기준 V-Q 대비 scale 축소 proxy. 절대 LAM% 아님; 부호·bound 포화에 주의. | 계산: (1 - fit_scale) * 100 (discharge 3-param fit). scale>1 이면 음수.

트렌드: LAM 곡선 proxy: early=-3.194 → late=-16.44 ($\Delta=-13.24$, -5.251%/100cyc) → decreasing · context | aging 기대=either | vs=context

LLI 곡선 proxy (LLI_curve_proxy)

기준 대비 Q축 offset. 절대 LLI% 아님. | 계산: fit_offset / Q_max × 100.

트렌드: LLI 곡선 proxy: early=-0.5851 → late=-2.774 ($\Delta=-2.189$, -1.046%/100cyc) → decreasing · context | aging 기대=either | vs=context

Rate capability factor (RCF)

0.5C 용량 / 가까운 C/3 RPT 용량. rate 손실 시 하락. | 계산: Q_0.5C(N) / Q_C/3(nearest RPT).

트렌드: Rate capability factor: early=0.9687 → late=0.9519 ($\Delta=-0.01679$, -0.0012021/100cyc) → flat · stable | aging 기대=decrease | vs=stable

지표 설명 · 트렌드 (M02Ch105)

분극 효율 비 (PER)

과전위 대비 DCIR 스케일. 수송 제한 시 변화. | 계산: $\eta(\text{SOC50}) / (\Delta I \cdot R_{\text{DCIR}_50})$.

트렌드: 데이터 부족 | aging 기대=increase | vs=n/a

최대 과전위 η (eta_max)

C/3 vs 0.5C 동일 Q속 최대 전압차. | 계산: 동일 Q에서 $|V_{\text{C3}} - V_{\text{0.5C}}|$ 최댓값.

트렌드: 데이터 부족 | aging 기대=increase | vs=n/a

η 최댓값 SOC (eta_argmax_SOC)

η 가 최대인 SOC. 고SOC면 PE, 저SOC면 NE 쪽 힌트. | 계산: argmax_SOC of $\eta(\text{SOC})$.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

PE activity 패턴 (LAM_PE_pattern_score)

NCM 이차 고립/activity 패턴. 절대 LAM% 금지. | 계산: plateau·dQdV·LAM_curve_proxy·dQV_log_var 가중합 (mode_weights).

트렌드: PE activity 패턴: early=0.6565 → late=0.9319 (Δ =+0.2754, +0.1380–1/100cyc) → increasing · matches_aging | aging 기대=increase | vs=matches_aging

접촉/스택 손실 (contact_loss_score)

옴 성장·분율·mech/chem 증가 증거 합. 전극 미분해. | 계산: R_ohmic_growth, $\Delta R\Omega$, $R\Omega_{\text{frac}}$, EoC_10s, mech/chem 가중합.

트렌드: 접촉/스택 손실: early=0.8053 → late=0.9259 (Δ =+0.1206, +0.074440–1/100cyc) → increasing · matches_aging | aging 기대=increase | vs=matches_aging

LLI 패턴 (LLI_pattern_score)

CE·slippage·곡선 offset 기반 LLI 가설 점수. | 계산: CE ↓, CI ↑, restV/cutoff margin, LLI_curve_proxy 등.

트렌드: LLI 패턴: early=0.3602 → late=0.6325 (Δ =+0.2723, +0.11260–1/100cyc) → increasing · matches_aging | aging 기대=increase | vs=matches_aging

지표 설명 · 트렌드 (M02Ch105)

계면 R 패턴 (interface_R_score)

Rct·τ·VE 하락 등 계면저항 성장 가설. | 계산: R_ct ↑, tau_ct ↑, VE ↓, EoC_60s 등.

트렌드: 계면 R 패턴: early=0.4692 → late=0.8449 ($\Delta=+0.3757$, $+0.15480-1/100\text{cyc}$) → increasing · matches_aging | aging 기대=increase | vs=matches_aging

고체확산 제한 (solid_diffusion_score)

A_diff·PER·RCF 기반 확산/수송 제한. | 계산: A_diff ↑, PER ↑, RCF ↓ 등.

트렌드: 고체확산 제한: early=0.555 → late=0.7449 ($\Delta=+0.1899$, $+0.036970-1/100\text{cyc}$) → flat · stable | aging 기대=increase | vs=stable

PE lean 점수 (PE_side_score)

LAM_PE_pattern + feature boost + FC-OCP Δhits. | 계산: $0.75 \cdot \text{LAM_PE} + 0.20 \cdot \text{PE_boost} + \text{peak_boost}$.

트렌드: PE lean 점수: early=0.5423 → late=0.7989 ($\Delta=+0.2566$, $+0.11390-1/100\text{cyc}$) → increasing · matches_aging | aging 기대=increase | vs=matches_aging

contact_stack lean (contact_stack_score)

contact_loss를 dominant 경쟁에 넣는 이름. $\approx \text{contact_loss_score}$. | 계산: $\text{clip}(\text{contact_loss_score}, 0, 1)$.

트렌드: contact_stack lean: early=0.8053 → late=0.9259 ($\Delta=+0.1206$, $+0.074440-1/100\text{cyc}$) → increasing · matches_aging | aging 기대=increase | vs=matches_aging

NE 가설 점수 (NE_side_score)

contact × Si chemo-mech co-sign. Si 없으면 NE 확정 금지. | 계산: $0.55 \cdot \text{contact} \cdot (0.25 + 0.75 \cdot \text{si}) + 0.30 \cdot \text{NE_boost} \cdot \text{si}$.

트렌드: NE 가설 점수: early=0.1772 → late=0.2804 ($\Delta=+0.1032$, $+0.055850-1/100\text{cyc}$) → increasing · matches_aging | aging 기대=increase | vs=matches_aging

Si co-sign (si_cosign)

저SOC hyst·Q_relax·mech/chem·CV 등 Si chemo-mech 동시 신호. | 계산: SI_NE_COSIGN feature boost (baseline 대비 ↑ 비율).

트렌드: Si co-sign: early=0.2 → late=0.4 ($\Delta=+0.2$, $+0.10130-1/100\text{cyc}$) → increasing · matches_aging | aging 기대=increase | vs=matches_aging